ASRAF SIDDIQUEE

South Bend, IN | 870-273-9430 | nayem.asraf@gmail.com | LinkedIn | Portfolio

Education

Ph.D. Candidate in Electrical Engineering [University of Notre Dame, Notre Dame, IN] Expected: May 2027
M.S. in Electrical Engineering [University of Notre Dame, Notre Dame, IN] 2023
Master of Engineering Management [Arkansas State University, Jonesboro, AR] 2022
M.S. in Mechanical Engineering [Arkansas State University, Jonesboro, AR] 2017
B.S. in Mechanical Engineering [Khulna University of Engineering & Technology, Bangladesh] 2013

Technical Skills

- **Mechanical Design & Product Development:** SolidWorks, Fusion 360 (Parametric & Generative), Siemens NX, AutoCAD, GD&T, Design for Manufacturing (DFM), Tolerance Analysis, Multi-Material Integration.
- **Rapid Prototyping & Fabrication:** 3D Printing (FDM/SLA), Laser Cutting, Resin Casting, Multimaterial Integration (Plastics, Elastomers, Fabrics, and Softgoods), Stitching, CNC Machining, Sheet Metal Fabrication, Stamping, Forging, and Sand Casting, Plastic Injection Molding.
- **Robotics & Mechatronics:** Proteus, KiCad, Arduino, ESP32, Raspberry Pi.
- **Simulation & Modeling:** Abaqus, COMSOL, ANSYS.
- **Programming:** Python, MATLAB, C++.

Industry Experience

Northern Electric Company, Inc.
South Bend, IN
Electric Motor Engineering Trainee (Part-Time)
October 2026 – Present

- **Assist experienced technicians** with electric motor assembly, maintenance, troubleshooting, repair, and testing through hands-on industrial training.
- **Develop practical expertise in electric motor components and operation**, including AC/DC motors, servo motors, stator windings, rotor assemblies, and encoders.

AceOne Technologies
Jonesboro, AR, USA
Hardware Engineer Intern
Jan. 2020 – Jul. 2021

- **Developed embedded C++/Arduino firmware** for real time multi sensor data acquisition.
- **Designed and validated PackRight device hardware**, including enclosure redesign and sensor integration.

A-State Innovate, The Garage
Jonesboro, AR, USA
Mechanical Engineer (Volunteer)
May. 2019 – Jan. 2020

- Designed and fabricated **four pneumatic grippers** with custom mounting mechanisms and multi-material integration.
- Designed and fabricated mechanical prototypes including a **multi cylinder mechanism, helical bevel gear**, stair structure, and table structure using CAD, additive manufacturing, GD&T, and DFAM

Adroit Engineers
Dhaka, Bangladesh
Mechanical Design & Prototyping Engineer
Oct. 2013 – Jan. 2015

- **Led mechanical design, FEA, and embedded system development (PCB, C/C++)** for electromechanical medical devices, including automated wheelchairs and hospital beds.
- **Optimized structural design** of automated wheelchair systems, achieving **7% reduction in weight and cost**.
- **Led mechanical product development** from design through prototyping and validation; developed an automated wheelchair with gearbox and linear-actuator systems and designed a gearbox mechanism for hospital-bed adjustment.

Research Experience

miNiRo Lab, University of Notre Dame — Notre Dame, IN, USA
Graduate Research Assistant — Aug. 2022 – Present

- **Hierarchical Differentiable Co-Design:** Developing a physics-informed machine learning framework to reduce swimmer body drag and leveraging differentiable simulation with PolyFEM to optimize soft pneumatic leg cavity geometry for faster, more energy-efficient locomotion.
- **Reconfigurable Soft Robotic Leg with Adaptive Flippers:** Designing and prototyping a reconfigurable soft robotic leg with passively adaptive flippers for terrestrial and aquatic locomotion, integrating compliant structures and modular mechanisms.
- **Led a 6-month DoD-funded project**, developing a fully operational **underwater burrowing robot** from concept through fabrication and experimental validation.
- **Designed and optimized bio-inspired soft robotic systems** for terrestrial, underwater, and surface-water locomotion, including multi-legged robots, **sea star-inspired robots**, and soft grippers.
- **Performed nonlinear FEA in Abaqus** on silicone pneumatic actuators, PLA structures, and compliant skins, evaluating deformation, stress, fluid-structure interaction (CEL), and anisotropic friction.
- **Integrated a Robotiq FT 300-S force/torque sensor** with a custom mount for robotic actuator force measurement, data acquisition, and experimental validation.

Thermal Engineering Lab, Arkansas State University — Jonesboro, AR, USA
Graduate & Postgraduate Research Assistant — Jan. 2015 – Aug. 2018

- Developed predictive models for ESP and SCR systems in an 870 MW supercritical power plant, modeling NO_x reduction and ash deposition.
- Contributed to $1.22M federally funded research; modeled VRFB systems (ASTA $79,379) and supported NSF EPSCoR energy storage test-bed development ($120,000).

Technical Projects

- **Edison Innovation Fellowship, Soft-Legged Surface Water Explorer:** Selected as 1 of 10 university-wide fellows (2026–2027, $17,077.50 stipend) to develop an autonomous soft-legged aquatic robot for environmental monitoring in shallow, vegetation-rich waterways.
- **Soft Rigid Hybrid Modular Terrestrial Robot**: Developing a hybrid soft rigid locomotor integrating a plastic skeletal frame, silicone compliant structure, and custom BLDC motor gearbox system for modular terrestrial locomotion.
- **Baja SAE eCVT Mechanical Design Project**: Leading a team of four undergraduate students in the design and development of an electronic continuously variable transmission (eCVT) for the University of Notre Dame Baja SAE vehicle.
- **ANDBot: Reconfigurable Amphibious Robot for Cross Environmental Exploration**: Designed a reconfigurable robotic platform for locomotion across terrestrial and aquatic environments.
- **Shape Adaptive Soft Gripper for Picking Thin and Small Objects**: Designed and developed a compliant soft gripper for grasping small, thin, and difficult to pick objects.
- **Wearable Soft Finger Rehabilitation Robot**: Developed a wearable pneumatic soft actuator system for controlled finger bending and rehabilitation assistance.

Selected Publications

Journal Articles

- **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. Design and Simulation of Soft, Multi-Legged Robotic Swimmer for Shallow Water Locomotion. [Under Review]
- **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. (2024). Utilizing Bioinspired Soft Modular Appendages for Grasping and Locomotion in Multi-Legged Robots on Ground and Underwater. IEEE Robotics and Automation Letters (RA-L).

Conference Papers

- Siddiquee, A. N. M. A., Castillo Gonzalez, M., Utama, V., Zhu, Y., & Ozkan-Aydin, Y. (2026). Design and Iterative Evaluation of a Sea Star-Inspired Robot with Autonomous Ray. **[Accepted: IEEE/RSJ IROS 2026]**
- **Siddiquee, A.**, & Jeong, K. (2016). Conjugated Dynamic Modeling on Vanadium Redox Flow Battery With Non-Constant Variance for Renewable Power Plant Applications. Proceedings of the ASME 2016 International Mechanical Engineering Congress and Exposition (IMECE2016-67462), Phoenix, AZ, USA.